

Anuprovo Debnath

✉ ad23ms110@iiserkol.ac.in

☎ +91 6289630036

🌐 <https://anuprovo-debnath.github.io/>

🌐 [LinkedIn](#)

Address: Sundaram Apartment, Flat-403, Harmoz Villa, Madhyamgram, Kolkata-700129, West Bengal, India

Education

- 2023 – Present  **BS-MS** Indian Institute of Science Education and Research, Kolkata
Major: Physics
Minor: Mathematics, Computer Science
CGPA: 8.82
- 2023  **I.S.C.** St. Jude's High School
Class 12 - 95.7%
- 2021  **I.C.S.E** St. Jude's High School
Class 10 - 94.4%

Projects

Numerical Modeling of Laser-Plasma Wakefield Acceleration

05.2026 – Present

Under the guidance of Prof. Sudip Sengupta (Basic Theory and Simulation, Institute for Plasma Research), developed a custom Python-based Runge-Kutta (RK4) computational framework to model electron trapping and acceleration dynamics in laser-driven plasma waves. Analytically derived and computationally verified the energy gain of test electrons in a longitudinal sinusoidal electric field ($E_0 \sin(kx - \omega t)$), successfully reproducing the theoretical energy scalings of Tajima and Dawson (1979). Extended the model to a fully vectorized 3D architecture with an applied perpendicular magnetic field to simulate the phase-matching conditions of the “Surfatron” acceleration regime (Katsouleas & Dawson, 1983). Currently building upon this theoretical foundation by developing a custom 1D Electrostatic Particle-in-Cell (PIC) simulation code from scratch to self-consistently model kinetic plasma instabilities and wave-particle interactions.

Simulation and Analysis of ns-DBD Plasma Actuator Shock Waves

06.2026 – 07.2026

As an IASc Summer Research Fellow under Dr. T. Murugan (CSIR-CMERI), engineered an OpenFOAM framework to simulate ns-DBD plasma actuator shock waves at 30, 70, and 100 kPa. By implementing a 100 ns phenomenological energy deposition model on an optimized multi-block grid, the computational overhead of plasma chemistry was successfully bypassed. Additionally, developed a custom Python/PyVista diagnostic pipeline utilizing Modified Friedlander Waveform fitting and Levenberg-Marquardt uncertainty propagation to extract sub-grid kinematics, successfully validating the numerical shock propagation against benchmark experimental data.

SRFP 8-week Internship Report: [Link](#)

Evaluating Plasma Parameters from a Typical LIBS Plasma

05.2025 – 10.2025

Under the guidance of Dr. Prashant Kumar (Scientist, PRL Ahmedabad), developed a Python-based framework for extracting electron temperature (T_e) and electron density (n_e) from Laser-Induced Breakdown Spectroscopy (LIBS) plasmas using Boltzmann plots, Saha-Boltzmann analysis, and Stark broadening with NIST spectral data. The validated approach demonstrates that calibration-free LIBS can reliably estimate plasma parameters. Currently extending the work by implementing a machine-learning framework with LTE validation, modelling self-absorption, and improving spectral line analysis.

Summer Internship Report: [Link](#)

Hobbyist Compute Cluster Setup (Slashdot Coding Club)

11.2025 – Present

As Secretary of the coding club *Slashdot*, leading the setup of a compute cluster using refurbished PCs sourced from various institute laboratories. Implementing virtualization and orchestration using Proxmox and Kubernetes to provide a sandboxed environment for experimentation without impacting the mission-critical compute cluster of IISER Kolkata.

Simulation of Accretion Disk Around a Black Hole

02.2025 – Present

Project under the supervision of Dr. Sudip K. Garain (Assistant Professor, DPS, IISER Kolkata). Simulating accretion disk dynamics around a black hole in a relativistic framework, following concepts from *Fluid Mechanics* (6th Edition) by Kundu, Cohen, and Dowling.

Reading Project: Thermodynamics and Statistical Mechanics

05.2024 – 07.2024

Reading project under Prof. Arindam Kundagrami (DPS, IISER Kolkata). Studying thermodynamics and statistical mechanics following '*A Treatise on Heat*' by M. N. Saha and B. N. Srivastava.

Skills

Coding	Java, Python, FORTRAN, C, C++, $\text{\LaTeX}$
Data Visualisation	gnuplot, Matplotlib, ORIGIN, ParaView
Web Dev	HTML, CSS, JavaScript
Misc.	Blender, $\text{\LaTeX}$ and Typst
Languages	Strong reading, writing and speaking competencies for English, Bengali and Hindi.

Experience

- 2025 **Qiskit Fall Fest 2025, Slashdot Coding Club**
Organized an [IBM-sponsored quantum computing event](#), bringing together students from across the country for hands-on learning using Qiskit. First-of-its-kind event organized by SlashDot.
- Blender Workshop on Scientific Visualisation**
Conducted a workshop on using Blender for scientific visualisation using Python scripting.
- 2024 **IISER Mohali Astronomy & Astrophysics Summer School**
Participated in the summer school, gaining hands-on experience in astronomy and astrophysics research techniques.
- SAGAN Workshop, Exoplanet Summer Hybrid Workshop**
Attended the NASA Exoplanet Science Institute-hosted workshop on exoplanet research.
- iGEM Team, IISER Kolkata**
Contributed to website development, visualisation and animation for our iGEM team.

Awards and Achievements

- 2026 **SRFP**
[Selected](#) for research in Physics at CSIR-CMERI as part of SRFP 2026 by IAS/INSA/NASI.
- 2025 **Smart India Hackathon (SIH)**
Our team '[Haklu El At](#)' was shortlisted in problem statement: SIH25173.
- 2024 **International Astronomy and Astrophysics Competition**
Received Silver Honour.
- 2023 **DST INSPIRE Scholarship**
For being in top 1% in ISC Examination.
- National Entrance Screening Test (NEST)**
All India Rank 48.
- 2021 **JBNSTS (Jagadis Bose National Science Talent Search) Junior Scholarship**
- KVPY (Kishore Vaigyanik Protsahan Yojana) Aptitude Test (Stream - SA)**
All India Rank 2104.
- 2016 – 2019 **Swarnakana Interschool Merit Test** organised by the **Rotary Club of Madhyamgram**
Ranked in every test from 2016 to 2019.

Extracurriculars

- Secretary of *Slashdot*, the coding and designing club of IISER-K, leading initiatives during the successful 2025–26 tenure.
- Active member of *Singularity*, the astronomy club at IISER-K. Served as a video editor and contributed significantly to the podcast [Talking to the Stars](#), featuring interviews with prominent astrophysicists across India.
- Active participant in theatrics, specializing in light and sound production. Part of the gold medal-winning drama team at the Inter INCI Cultural Meet (IICM) during 2024 and 2025.
- Assisted Prof. Rajesh Kumble Nayak, DPS, IISER-K in setting up a hobbyist electronics lab at IISER-K.
- Passionate about playing guitar and listening to music.